ENGINEERING PROJECT TECHNICAL SPECIFICATION & VIVA AUDIT

SmartAttend: Smart QR Attendance System

GPS Geofencing Proximity Verification & Anti-Proxy Security Engine

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ACADEMIC TERM
AY 2026–2027

1. EXECUTIVE SYSTEM ARCHITECTURE AUDIT

An end-to-end static code inspection and configuration analysis was conducted across all files, dependencies, algorithms, and modules of the **SmartAttend** repository.

Forensic Audit Finding: Architecture & Persistence Reality

SmartAttend is built as a **100% Client-Side Single Page Application (SPA)** powered by an **Embedded Geofence Rule Engine**. There is **no external backend server (Node.js/Express, Python/Django, Java/Spring)** and **no external SQL/NoSQL cloud database** connected. All authentication, session expiry, PRN uniqueness verification, and GPS proximity calculations execute in real-time in the browser runtime and persist via the `Web Storage API (localStorage)`.

🏛️ System Model: Client-Side SPA

- **Framework:** React 18.3.1 + React Router DOM 6.28.0
- **Bundler & Dev Server:** Vite 6.0.1 (ESM modules)
- **State Architecture:** React Context API (AttendanceContext & AuthContext)
- **Testing:** Automated Node.js Unit Suite (18 assertions)

🛡️ Anti-Proxy Geofencing Engine

- **Distance Engine:** Pure Haversine Great-Circle calculation ($R = 6,371,000 ext{m}$)
- **Hardware Sensor:** Native W3C Geolocation API (HTML5)
- **Accuracy Defense:** Rejection of inaccurate fixes (> ±50m)
- **Radius Enforcer:** Per-classroom boundary constraint (15m–35m)

2. EXHAUSTIVE TECHNOLOGY & TOOLING INVENTORY

TECHNOLOGY	CATEGORY	STATUS	ROLE IN SMARTATTEND CODEBASE
JavaScript (ES2022+)	Language	ACTIVELY IMPLEMENTED	Core programming language powering all components, context state machines, mathematical utilities, and tests.
HTML5 & Semantic DOM	Markup / Web API	ACTIVELY IMPLEMENTED	Document skeleton, HTML5 `<canvas>` rendering target for QR codes, file input listeners, and tables.
Vanilla CSS3	Styling System	ACTIVELY IMPLEMENTED	Handcrafted design system using CSS Variables (`variables.css`), Flexbox, CSS Grid, and Keyframes. Zero CSS framework overhead.
React 18.3.1	Frontend Framework	ACTIVELY IMPLEMENTED	Declarative UI with Hooks: `useState` , `useEffect` , `useRef` , `useMemo` , `useCallback` , `useContext` .
React Router DOM 6.28.0	Client Routing	ACTIVELY IMPLEMENTED	SPA navigation, route guards (`ProtectedRoute.jsx`), URL token parameter parsing (`useSearchParams`).
W3C Geolocation API	Sensor API	ACTIVELY IMPLEMENTED	Hardware sensor access via `navigator.geolocation.getCurrentPosition` with high-accuracy GPS fixes on student & teacher devices.
Haversine Distance Engine	Algorithm	ACTIVELY IMPLEMENTED	Spherical trigonometry function in `geoUtils.js` computing surface distance between student coords and registered room coords.
qrcode (v1.5.4)	QR Generation	ACTIVELY IMPLEMENTED	Draws dynamic session QR codes on HTML5 canvas with error correction level 'H' (30% recovery) and downloadable PNG export.
html5-qrcode (v2.3.8)	QR Scanner	ACTIVELY IMPLEMENTED	Real-time camera feed reader using rear environment camera, with local image file upload parsing fallback.
recharts (v2.13.3)	Data Visualization	ACTIVELY IMPLEMENTED	Interactive responsive charts in `Reports.jsx` : Weekly Attendance Bar Chart and Subject-Wise Turnout Line Chart.
lucide-react (v1.16.0)	Iconography	ACTIVELY IMPLEMENTED	SVG icons across portals: `QrCode` , `MapPin` , `Navigation` , `ShieldCheck` , `Camera` , etc.
canvas-confetti (v1.9.4)	Animation	ACTIVELY IMPLEMENTED	Celebratory confetti particle blast rendered upon verified physical classroom attendance.
Web Storage API	Data Persistence	ACTIVELY IMPLEMENTED	Acts as the client database. Stores classrooms (`classrooms_v2`), active sessions (`session_v3`), and logs (`attendance_v3`).
Vite (v6.0.1)	Build Tool / Bundler	ACTIVELY IMPLEMENTED	Modern fast build environment and local HTTP dev server configured in `vite.config.js` .
Node.js Test Suite	Testing Framework	ACTIVELY IMPLEMENTED	Automated unit tests in `tests/geofence_verification.test.js`

TECHNOLOGY	CATEGORY	STATUS	ROLE IN SMARTATTEND CODEBASE
			validating 18 mathematical and edge-case assertions.

3. TECHNOLOGY REALITY & CLASSIFICATION MATRIX

Distinguishing clearly between what is actively running, what is installed, and what is absent is essential for project defense integrity:

STATUS CATEGORY	TECHNOLOGIES & COMPONENTS	VIVA DEFENSE GUIDANCE
ACTIVELY IMPLEMENTED & RUNNING	React 18, Vite 6, React Router DOM v6, Haversine Engine, W3C Geolocation, qrcode, html5-qrcode, recharts, lucide-react, canvas-confetti, localStorage, Node unit tests.	Showcase these directly in the live software demo. Every feature has working UI and logic.
DEPENDENCIES IN PACKAGE.JSON	All 7 runtime dependencies (`canvas-confetti` , `html5-qrcode` , `lucide-react` , `qrcode` , `react` , `react-dom` , `react-router-dom` , `recharts`) are actively imported.	Zero dead dependencies. 100% utilization rate of all installed npm packages.
NOT FOUND / NOT COMMITTED	• Express.js / Node.js Server API • MongoDB / PostgreSQL / MySQL • Firebase / Supabase • WebSockets (Socket.io) • TypeScript	**Do NOT claim in Viva:** Frame the system as a high-security Single Page Application with an Embedded Geofence Engine rather than claiming an uncommitted backend.

4. GPS GEOFENCING & HAVERSINE DISTANCE ENGINE

Standard QR systems fail because a student in a canteen can easily scan a QR code screenshot forwarded over WhatsApp. SmartAttend eliminates this with a **two-layer physical verification handshake**.

A. The Haversine Spherical Distance Equation

Because the Earth is an oblate spheroid, simple Euclidean distance ($sqrt{Delta x^2 + Delta y^2}$) causes massive geographic distortion. SmartAttend calculates surface distance across the globe using the **Haversine Formula**:

```
Δφ = lat₂ - lat₁ (in radians) | Δλ = lon₂ - lon₁ (in radians)
a = sin²(Δφ / 2) + cos(φ₁) · cos(φ₂) · sin²(Δλ / 2)
c = 2 · atan2(√a, √(1 - a))
d = R · c [where Earth's Mean Radius R = 6,371,000 meters]
```

Calculated in `src/utils/geoUtils.js` with precision rounded to **0.1 meters**:

```
export const calculateHaversineDistance = (lat1, lon1, lat2, lon2) => {
  const phi1 = toRadians(lat1), phi2 = toRadians(lat2);
  const deltaPhi = toRadians(lat2 - lat1), deltaLambda = toRadians(lon2 - lon1);
  const a = Math.sin(deltaPhi / 2)**2 + Math.cos(phi1) * Math.cos(phi2) * Math.sin(deltaLambda / 2)**2;
  const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
  return Math.round(EARTH_RADIUS_METERS * c * 10) / 10;
};
```

B. Sensor Accuracy Safeguard (Anti-Spoofing & Weak Fix Filter)

> **Why GPS Accuracy Threshold is Mandatory:**
> A mobile phone deep indoors often switches from satellite GPS to cellular cell-tower triangulation, giving an inaccurate position with an accuracy radius of `±80m to ±150m`. If unhandled, this could falsely mark an absent student present. SmartAttend evaluates `position.coords.accuracy`: if accuracy exceeds the classroom threshold (default `±50m`), attendance is blocked with `POOR_GPS_ACCURACY`, instructing the student to step near a window.

5. LIVE DEMONSTRATION PLAYBOOK (VIVA PRESENTATION GUIDE)

During external evaluation, demonstrate the system's resilience using the built-in demo presets on `/student/details`:

DEMO SCENARIO	TEST BUTTON IN UI	SYSTEM BEHAVIOR	WHAT TO EXPLAIN TO EXAMINERS
1. Physical Presence Verified	🟢 Inside Class	Distance: **4.5m** (≤ 20m) Result: **Success + Confetti**	"The student's coordinates match Room 204. Proximity is validated, Present record is logged with timestamp and GPS accuracy."
2. Proxy Attempt Blocked	🔴 In Canteen	Distance: **135m** (> 20m) Result: **OUT_OF_BOUNDS Rejection**	"A proxy attempt from the college canteen or hostel is instantly detected. The red error screen displays distance and rejects attendance."
3. Weak Sensor Fix Filter	🟡 Weak GPS	Accuracy: **±95m** (> ±50m) Result: **POOR_GPS_ACCURACY**	"Demonstrates sensor accuracy validation: Weak cellular fixes are rejected to prevent false positive attendance records."
4. Real Device Sensor	🔵 Live Device GPS	Reads real laptop/phone GPS sensor live	"Demonstrates live integration with W3C Geolocation API reading device hardware latitude and longitude coordinates."

6. SENIOR ENGINEER VIVA VOCE DEFENSE GUIDE

Q1: Why did you use the Haversine formula instead of the standard distance formula (Pythagorean Theorem)?

Model Answer: "The Earth is a curved sphere, not a flat Euclidean plane. The standard Pythagorean formula $sqrt{Delta x^2 + Delta y^2}$ assumes flat coordinates and introduces significant distortion over latitude degrees. The Haversine formula calculates the great-circle spherical distance across the Earth's radius ($R = 6,371,000 ext{m}$), ensuring meter-accurate precision."

Q2: How does your system prevent proxy attendance if someone shares a picture of the QR code?

Model Answer: "Even if a student forwards a photograph of the QR code to an absent friend via WhatsApp, scanning the code only initiates the verification process. The student cannot submit attendance without providing their live GPS coordinates. When evaluated against the registered classroom geofence, any student beyond the permitted 20m radius is blocked with an `OUT_OF_BOUNDS` security violation."

Q3: How do you handle QR code expiration and session security?

Model Answer: "Every QR code is generated with a time-bounded cryptographic session token (e.g. `DEMO_CSF_02`) and an expiry timestamp (1, 2, 5, or 10 minutes). Once the countdown timer reaches zero, the QR code automatically grays out, the token is deactivated, and any subsequent submissions are rejected with an `EXPIRED` error."

Q4: What stops a student from submitting multiple attendances with the same PRN?

Model Answer: "In `AttendanceContext.jsx`, before creating any attendance record, the system cross-checks the student's PRN against the existing records for the current session ID and date. If a record already exists, it is blocked with an `ALREADY_MARKED` constraint."

Q5: How does the camera scanner handle poor lighting or older phone cameras?

Model Answer: "We built a multi-modal scanning architecture in `ScannerComponent.jsx`: (1) Live camera scanner with `html5-qrcode` targeting the rear camera, (2) Image file upload fallback allowing students to select a captured QR image, and (3) Manual session token input for faculty-assisted entry."

Q6: Where is attendance stored, and how can teachers export reports?

Model Answer: "Attendance records are managed in local application state and synced to `localStorage`. Faculty can monitor attendees in real time and download a full CSV audit report containing PRN, student name, verification timestamp, GPS distance, and geofence verification status with a single click."

Q7: How did you test and verify your geofencing algorithm?

Model Answer: "We wrote an automated unit testing suite in `tests/geofence_verification.test.js` containing 18 assertions. It runs in Node.js and tests inside-room cases (5m), out-of-bounds cases (140m), exact boundary cases (20m), weak GPS accuracy cases (±95m), coordinate sanitization, and mathematical precision."

Q8: What are the future production extensions of this project?

Model Answer: "In future production phases, the embedded geofence verification logic can be migrated to a backend microservice (Node.js/Express) with JWT token signing, an encrypted PostgreSQL database, and WebSocket listeners for real-time lecture hall dashboards."
